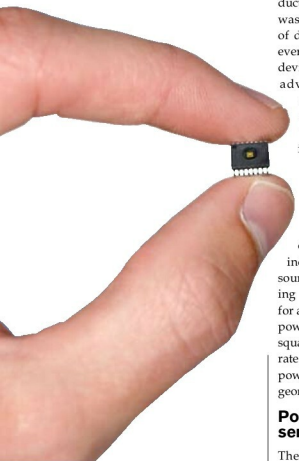


LOWER POWER CONSUMPTION THROUGH VLSI DESIGN

Depending upon the architecture of the design, the designers can choose from a wide range of options to reduce power consumption of VLSI circuits. Take a look at some of them



ductor chip designers in the late 1980's was area constraint, i.e., optimisation of design to reduce chip area. However, with continuous scaling of the device to minimise feature size and advancement in computer-aided design tools, area constraint has faded away. Tens of millions of transistors can now be fabricated in a single chip.

Today, power and performance have become the predominant concerns for chip designers. Low power consumption is vital for handheld mobile and wireless devices as most of them use an independent battery as the power source. Power management is becoming an increasingly urgent problem for almost every category of design, as power density (measured in watts per square millimetre) rises at an alarming rate. Fig. 1 shows an increase in the power density with shrinking device geometry.

Power dissipation in a semiconductor device

The power dissipation in a CMOS semiconductor device is classified as

static and dynamic. The static power dissipation is due to reverse saturation, sub-threshold and leakage current and occurs especially when the device is in idle mode. The dynamic power dissipation occurs due to charging and discharging of load capacitance and short-circuit current when both the NMOS and PMOS devices remain 'on' for a short time duration:

$$P_{\text{switching}} = \alpha f C_{\text{eff}} V_{\text{dd}}^2$$

where 'α' is switching activity, 'f' is switching frequency, C_{eff} is effective capacitance and V_{dd} is supply voltage.

$$P_{\text{short-circuit}} = I_{\text{sc}} V_{\text{dd}} f$$

where I_{sc} is short-circuit current during switching, V_{dd} is supply voltage and 'f' is switching frequency.

Dynamic power can be lowered by reducing the switching activity and clock frequency, which affects performance, and also by reducing the capacitance and supply voltage.

Leakage power is a function of supply voltage V_{dd} , switching threshold voltage V_{th} and the transistor size:

$$P_{\text{Leakage}} = f(V_{\text{dd}}, V_{\text{th}}, W/L)$$

where V_{dd} is supply voltage, V_{th} is threshold voltage, 'W' is transistor width and 'L' is transistor length.

■ RITU RAJ LAMSAI

"If the automobile industry advanced as rapidly as the semiconductor industry, a Rolls Royce would get half a million miles per gallon and it would be cheaper to throw it away than to park it"

— Gordon Moore, 1998

Semiconductors have seen a revolution over a decade and the trend will continue in terms of area, speed and power.

The major worry for the semicon-

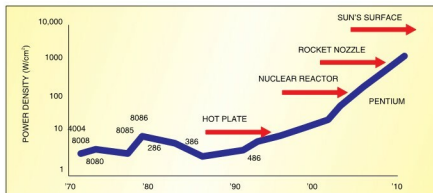


Fig. 1: Power density with shrinking geometry